

FortTrace Plant Operations
Dahej Integrated Manufacturing Complex, Bharuch District, Gujarat - 392130

**QUARTERLY ENVIRONMENTAL COMPLIANCE
REPORT**

Quarter 2, Financial Year 2024-25 (April - June 2024)

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Facility	FortTrace Dahej Integrated Manufacturing Complex (Unit-1 & Unit-2)
Consent to Operate No.	GPCB/CTO/BHR-DAHEJ/2022/4417
Regulatory Authority	Gujarat Pollution Control Board (GPCB) / CPCB
Prepared By	EHS & Sustainability Department
Reviewed By	Head - Environment, Health & Safety
Approved By	Plant Manager, FortTrace Dahej Complex
Overall Compliance Status	COMPLIANT (GREEN)
Date of Issue	08-Jul-2024

This report is generated in compliance with the Consent to Operate conditions issued under the Water (Prevention & Control of Pollution) Act, 1974, the Air (Prevention & Control of Pollution) Act, 1981, and the Environment (Protection) Act, 1986, and is submitted for internal record-keeping and regulatory audit readiness.

Document Control & Revision History

Rev	Date	Description	Prepared By	Approved By
00	08-Jul-2024	Initial issue - Q2 FY2024-25 compliance report	A. Deshmukh, Sr. Executive - EHS	R. Nair, Plant Manager

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List of Abbreviations

Abbreviation	Full Form
AAQ	Ambient Air Quality
BOD	Biochemical Oxygen Demand
CETP	Common Effluent Treatment Plant
COD	Chemical Oxygen Demand
CPCB	Central Pollution Control Board
CTE	Consent to Establish
CTO	Consent to Operate
DG Set	Diesel Generator Set
EC	Environmental Clearance
EHS	Environment, Health & Safety
ETP	Effluent Treatment Plant
GIDC	Gujarat Industrial Development Corporation
GPCB	Gujarat Pollution Control Board
HW	Hazardous Waste
KLD	Kilolitres per Day
Leq	Equivalent Continuous Sound Level
MoEFCC	Ministry of Environment, Forest and Climate Change
NAAQS	National Ambient Air Quality Standards
NABL	National Accreditation Board for Testing and Calibration Laboratories
NOx	Oxides of Nitrogen
OCMMS	Online Consent Management & Monitoring System
PM / PM10 / PM2.5	Particulate Matter (and size fractions, in microns)
SBR	Sequencing Batch Reactor
SO2	Sulphur Dioxide
TPH	Tonnes per Hour
TSDF	Treatment, Storage and Disposal Facility
TSS	Total Suspended Solids

1. Executive Summary

This report presents the environmental performance and regulatory compliance status of FortTrace Plant Operations' Dahej Integrated Manufacturing Complex for the second quarter of Financial Year 2024-25 (April to June 2024). Monitoring activities covered stack emissions, ambient air quality, effluent quality at the Effluent Treatment Plant (ETP) outlet, hazardous waste generation and disposal, and ambient noise levels across the facility boundary and process areas.

All parameters monitored during the reporting quarter were within the limits prescribed under the facility's Consent to Operate (CTO) issued by the Gujarat Pollution Control Board (GPCB), and within the National Ambient Air Quality Standards (NAAQS), 2009 and the applicable effluent and stack emission standards notified under the Environment (Protection) Rules, 1986. No exceedances requiring escalation to the State Pollution Control Board were recorded during this quarter.

Key Highlights

- Overall compliance status for Q2 FY2024-25: **GREEN (Compliant)**.
- Stack emissions from Boiler Stack SA-01 and DG Set Stack SA-02 remained within CTO limits across all 12 weekly monitoring cycles.
- Ambient air quality at all four monitoring stations (AAQ-1 to AAQ-4) remained within NAAQS limits for PM10 and PM2.5.
- ETP treated effluent quality met discharge norms for COD, BOD, TSS and pH at outlet point EF-01 for all monthly monitoring events.
- Hazardous waste generation was tracked and disposed through GPCB-authorized recyclers/TSDF in accordance with the Hazardous and Other Wastes (Management & Transboundary Movement) Rules, 2016.
- Two minor observations were recorded during internal EHS walkthroughs (Section 9); both were closed within the quarter with no regulatory implication.

2. Facility & Regulatory Overview

FortTrace Plant Operations' Dahej Integrated Manufacturing Complex is engaged in specialty chemical processing and downstream manufacturing operations across two production units (Unit-1 and Unit-2), supported by a captive power block, effluent treatment plant, and hazardous waste storage facility. The facility operates under a consolidated Consent to Operate issued by the Gujarat Pollution Control Board and is registered under the Environment (Protection) Act, 1986.

Parameter	Details
Facility Name	FortTrace Dahej Integrated Manufacturing Complex
Location	Plot No. GIDC-247/A, Dahej Industrial Estate, Bharuch District, Gujarat - 392130
Industry Category	Red Category - Specialty Chemicals & Process Manufacturing
Consent to Operate No.	GPCB/CTO/BHR-DAHEJ/2022/4417
CTO Validity	01-Apr-2022 to 31-Mar-2027
Consent to Establish No.	GPCB/CTE/BHR-DAHEJ/2020/1183
EC Reference (MoEFCC)	J-11011/104/2019-IA II(I)
Hazardous Waste Authorization No.	GPCB/HWM/BHR-DAHEJ/2023/0762
Reporting Units Covered	Unit-1 (Process Block A & B), Unit-2 (Process Block C), Captive Power Block, ETP, HW Storage Yard
Regulatory Authority	Gujarat Pollution Control Board (GPCB), Regional Office - Bharuch; Central Pollution Control Board (CPCB)

Monitoring under this report was conducted by the in-house EHS monitoring team using calibrated instrumentation (Annexure A) and cross-verified through third-party stack and ambient air monitoring conducted by MoEFCC-recognized laboratory, EnviroCheck Labs Pvt. Ltd. (NABL Accreditation No. TC-6821), once during the quarter as per CTO condition 14(b).

3. Stack Emission Monitoring

3.1 Monitoring Methodology & Instrumentation

Stack emissions were monitored weekly at two point sources: Boiler Stack SA-01 (attached to the 12 TPH coal-fired boiler, Unit-1) and DG Set Stack SA-02 (attached to the 1500 kVA standby diesel generator set). Sampling was carried out in accordance with CPCB-prescribed isokinetic sampling methods for Particulate Matter (IS 11255 Part 1) and instrumental analysis for SO2 (IS 11255 Part 2) and NOx (IS 11255 Part 7) using a continuous stack monitoring system supplemented by manual verification.

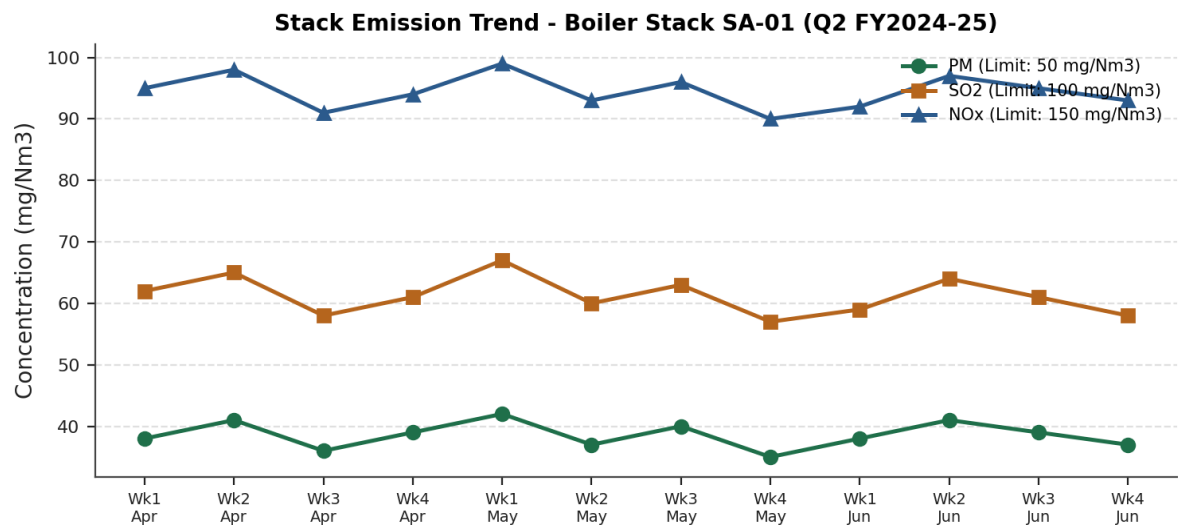
3.2 Stack Emission Results - Q2 FY2024-25 (Quarterly Average)

Stack ID	Source	Parameter	CTO Limit	Qtr. Avg.	Min	Max	Status
SA-01	Boiler (12 TPH, Coal-fired)	PM (mg/Nm3)	50	38.6	35	42	Compliant
SA-01	Boiler (12 TPH, Coal-fired)	SO2 (mg/Nm3)	100	61.3	57	67	Compliant
SA-01	Boiler (12 TPH, Coal-fired)	NOx (mg/Nm3)	150	94.4	90	99	Compliant
SA-02	DG Set (1500 kVA, Standby)	PM (mg/Nm3)	75	41.2	37	46	Compliant
SA-02	DG Set (1500 kVA, Standby)	SO2 (mg/Nm3)	-	N/A*	-	-	Not Applicable
SA-02	DG Set (1500 kVA, Standby)	NOx (mg/Nm3)	710 (as CO, ppm ref.)	312	289	334	Compliant

*SO2 not monitored for DG stack per CTO condition; DG set operates on HSD with sulphur content below prescribed threshold. NOx for DG stack reported against equivalent CPCB DG emission norm.

3.3 Stack Emission Trend Analysis

Weekly monitoring data for Boiler Stack SA-01 over the reporting quarter is presented below. All readings remained well within respective CTO limits with no exceedances recorded.



Marginal week-on-week variation is attributed to fluctuation in coal moisture content and boiler load factor; no corrective intervention was required as all values remained below 85% of the prescribed limit throughout the quarter.

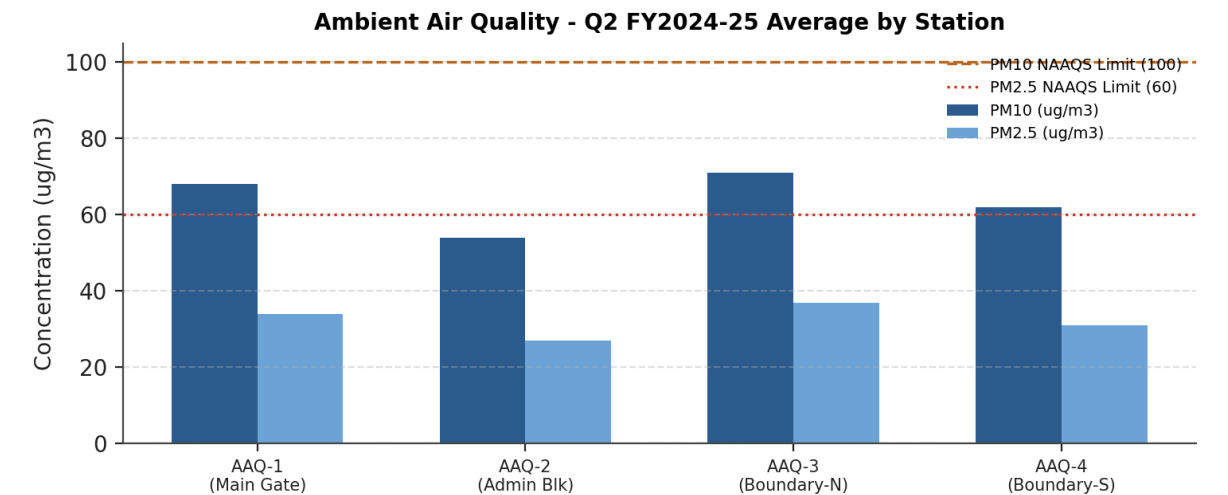
4. Ambient Air Quality Monitoring

4.1 Monitoring Station Details

Station ID	Location	Category	Frequency
AAQ-1	Main Gate / Security Post	Sensitive - Public Access	Twice weekly (24-hr avg.)
AAQ-2	Administrative Block	General - Occupational	Twice weekly (24-hr avg.)
AAQ-3	Boundary Wall - North (Downwind)	Boundary - Downwind	Twice weekly (24-hr avg.)
AAQ-4	Boundary Wall - South (Upwind)	Boundary - Upwind (Reference)	Twice weekly (24-hr avg.)

4.2 Ambient Air Quality Results (Quarterly Average)

Station	PM10 (ug/m3)	PM2.5 (ug/m3)	SO2 (ug/m3)	NOx (ug/m3)	CO (mg/m3)	Status
AAQ-1	68	34	12.4	26.1	0.8	Compliant
AAQ-2	54	27	9.8	21.4	0.6	Compliant
AAQ-3	71	37	14.1	28.7	0.9	Compliant
AAQ-4	62	31	10.6	23.5	0.7	Compliant
NAAQS Limit (2009)	100	60	80	80	2.0	-



5. Water Consumption & Effluent Quality

5.1 Water Consumption Summary

Source	Permitted Draw (KLD)	Actual Avg. Consumption (KLD)	% of Permitted
GIDC Water Supply (Industrial)	1,200	986	82.2%
Captive Bore-well (Non-potable use)	150	112	74.7%
Recycled/Treated Water (ETP Reuse)	-	318	-

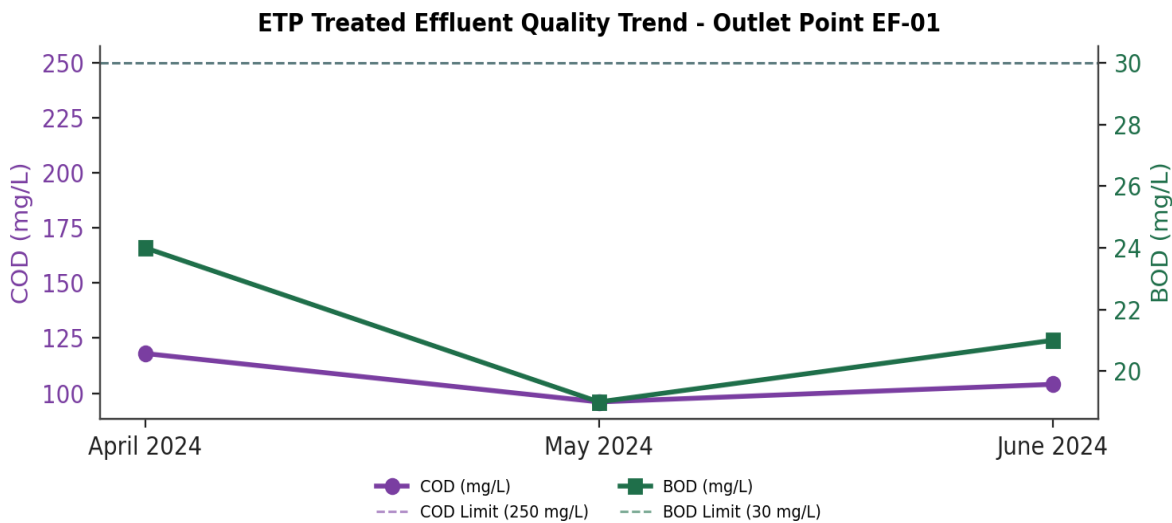
KLD = Kilolitres per Day. Recycled water reused for cooling tower make-up and horticulture, reducing net freshwater dependency.

5.2 Effluent Treatment Plant (ETP) Performance

The facility operates a physico-chemical + biological (SBR-based) Effluent Treatment Plant with a design capacity of 450 KLD, treating trade effluent from Unit-1 and Unit-2 prior to discharge to the GIDC Common Effluent Treatment Plant (CETP), Dahej, via the designated outlet point EF-01, in accordance with CTO discharge norms.

Parameter	Unit	CTO Limit (Inlet to CETP)	Apr-24	May-24	Jun-24	Status
pH	-	5.5 - 9.0	7.4	7.6	7.3	Compliant
TSS	mg/L	600	142	128	136	Compliant
BOD (3 days, 27C)	mg/L	30	24	19	21	Compliant
COD	mg/L	250	118	96	104	Compliant
Oil & Grease	mg/L	20	6.2	5.4	5.9	Compliant

5.3. Effluent Quality Trend Analysis



COD and BOD levels at the ETP outlet remained consistently below CTO discharge limits throughout the quarter, with a marginal reduction observed in May 2024 following optimization of the SBR aeration cycle. No instances of non-compliant discharge were recorded, and no discharge stoppage or bypass events occurred during the reporting period.

6. Hazardous & Solid Waste Management

Hazardous waste generated from process operations is stored in the designated Hazardous Waste (HW) storage yard under authorization GPCB/HWM/BHR-DAHEJ/2023/0762 and disposed through GPCB-authorized recyclers and the Bharuch Enviro Infrastructure Ltd. (BEIL) TSDF facility, in accordance with the Hazardous and Other Wastes (Management & Transboundary Movement) Rules, 2016.

Waste Category	HW Schedule Ref.	Qty Generated (MT, Q2)	Qty Disposed (MT, Q2)	Disposal Route
Spent Solvent	Sch-I, Cat 20.1	18.4	18.4	Co-processing - Authorized Cement Kiln
ETP Sludge	Sch-I, Cat 35.3	24.6	24.6	TSDF Landfill - BEIL Ankleshwar
Used/Spent Oil	Sch-I, Cat 5.1	3.2	3.2	Registered Re-refiner (Recycling)
Discarded Containers/Barrels	Sch-I, Cat 33.1	1.8	1.8	Authorized Recycler - Decontaminated
Chemical Sludge (Process)	Sch-I, Cat 26.1	9.1	8.6	TSDF Landfill - BEIL Ankleshwar (0.5 MT in transit)

All hazardous waste manifests (Form 10) were filed through the GPCB Online Consent Management & Monitoring System (OCMMS) within prescribed timelines. The 0.5 MT balance of Chemical Sludge generated in June 2024 was in transit to BEIL Ankleshwar as of the report cut-off date and was dispatched under manifest no. HW-DAHEJ-0629-114 on 02-Jul-2024.

Non-Hazardous / Solid Waste Summary

Category	Qty (MT, Q2)	Disposal Route
Scrap Metal	12.3	Authorized Scrap Vendor - Recycling
General/Domestic Waste	8.7	Municipal Solid Waste Collection - GIDC
Packaging Waste (Non-hazardous)	6.1	Authorized Recycler

7. Ambient Noise Level Monitoring

Ambient noise levels were monitored monthly at four locations representative of the facility boundary and occupational zones, in accordance with the Noise Pollution (Regulation and Control) Rules, 2000, for an Industrial Area classification.

Location	Day Limit dB(A) Leq	Night Limit dB(A) Leq	Day Avg. (Q2)	Night Avg. (Q2)	Status
Main Gate	75	70	62.4	54.1	Compliant
Boiler House Periphery	75	70	68.9	60.3	Compliant
Boundary Wall - North	75	70	58.2	49.7	Compliant
Boundary Wall - South	75	70	56.8	48.2	Compliant

Industrial Area limits as per Noise Pollution (Regulation and Control) Rules, 2000: Day (6:00-22:00 hrs) 75 dB(A) Leq; Night (22:00-6:00 hrs) 70 dB(A) Leq. Marginally elevated readings at Boiler House Periphery are attributable to induced draft fan operation and remain within prescribed limits.

8. Compliance Status Summary

The consolidated compliance status across all monitored environmental parameters for Q2 FY2024-25 is summarized below. This summary reflects the status displayed on the facility EHS Compliance Dashboard.

Compliance Area	Parameters Tracked	Exceedances	Status
Stack Emissions (SA-01, SA-02)	5	0	GREEN
Ambient Air Quality (AAQ-1 to AAQ-4)	5 per station	0	GREEN
Effluent Quality (EF-01)	5	0	GREEN
Hazardous Waste Management	Manifest & Storage Compliance	0	GREEN
Ambient Noise Levels	4 locations	0	GREEN
Water Consumption vs. Permitted Draw	2 sources	0	GREEN

Overall Compliance Status for Q2 FY2024-25: GREEN (COMPLIANT)
No exceedances of CTO conditions, NAAQS limits, or discharge norms were recorded during the reporting quarter. No show-cause notices, directions, or closure notices were received from GPCB or CPCB during this period.

9. Observations, Non-Conformances & Corrective Actions

The following minor observations were recorded during routine internal EHS walkthroughs conducted during the quarter. None of these observations resulted in a breach of CTO conditions or regulatory limits and are recorded here for completeness of the internal compliance record.

Obs. No.	Date	Observation	Corrective Action	Closure Date	Status
OBS-Q2-01	14-May-2024	Minor housekeeping lapse observed near HW storage yard - empty drums not stacked as per SOP.	Drums restacked; toolbox talk conducted with HW yard handling crew.	16-May-2024	Closed
OBS-Q2-02	03-Jun-2024	Continuous stack monitoring system data logger at SA-01 showed a 40-minute data gap due to local power fluctuation.	Logger reset; manual readings recorded during gap period and cross-verified; UPS backup checked and confirmed functional.	04-Jun-2024	Closed

No regulatory inspections were conducted at the facility by GPCB or CPCB during Q2 FY2024-25. The most recent joint inspection was conducted on 11-Feb-2024, with all findings closed prior to the current reporting period.

10. Annexure A - Monitoring Instrumentation & Calibration Status

Instrument	Make / Model	Serial No.	Last Calibration	Next Due	Calibrated By
Stack Sampler (Isokinetic)	Envirotech APM 620	ET-APM-3391	18-Feb-2024	17-Aug-2024	EnviroCheck Labs Pvt. Ltd.
Flue Gas Analyzer	Testo 350 XL	TS350-88214	22-Mar-2024	21-Sep-2024	Testo Calibration Center, Vadodara
Respirable Dust Sampler	Envirotech APM 460 NL	ET-APM-2207	05-Apr-2024	04-Oct-2024	EnviroCheck Labs Pvt. Ltd.
Fine Particulate Sampler (PM2.5)	Envirotech APM 550	ET-APM-5514	05-Apr-2024	04-Oct-2024	EnviroCheck Labs Pvt. Ltd.
Sound Level Meter	Larson Davis SoundAdvisor 831	LD-831-6602	10-Jan-2024	09-Jul-2024	Larson Davis Authorized Service, Mumbai
pH / Multiparameter Water Analyzer	Hach HQ40d	HQ40-19837	28-Feb-2024	27-Aug-2024	Hach India Service Center
BOD Incubator	Remi CIB-101	RMI-CIB-0245	12-Jan-2024	11-Jul-2024	In-house ETP Lab (NABL Traceable)
COD Digester / Titration Setup	Remi COD-24 Block Digester	RMI-COD-1129	12-Jan-2024	11-Jul-2024	In-house ETP Lab (NABL Traceable)

All instruments used for statutory monitoring are calibrated at intervals not exceeding six months against NABL-traceable reference standards. Calibration certificates are maintained in the EHS Document Portal under file reference FTP-EHS-CAL-2024.

Annexure B - Applicable Regulatory Standards Reference

The following regulatory standards and CTO conditions formed the basis for compliance evaluation presented in this report.

Environmental Medium	Governing Regulation	Applicable Standard
Stack Emissions	Air (Prevention & Control of Pollution) Act, 1981; CTO Schedule-II	Industry/fuel-specific limits per CTO (Section 3.2)
Ambient Air Quality	Environment (Protection) Rules, 1986	National Ambient Air Quality Standards (NAAQS), 2009 - CPCB Notification, 18-Nov-2009
Trade Effluent Discharge	Water (Prevention & Control of Pollution) Act, 1974; CTO Schedule-III	GPCB-prescribed inlet-to-CETP standards (Section 5.2)
Hazardous Waste Management	Hazardous and Other Wastes (Management & Transboundary Movement) Rules, 2016	Schedule-I waste categorization & authorized disposal routes
Ambient Noise	Noise Pollution (Regulation and Control) Rules, 2000	Industrial Area: Day 75 dB(A) Leq / Night 70 dB(A) Leq
Environmental Clearance Conditions	EIA Notification, 2006 (as amended)	EC No. J-11011/104/2019-IA II(I) - specific conditions

Where CTO-specific limits are more stringent than the general statutory standard, the CTO condition has been applied as the compliance benchmark throughout this report.

11. Certification & Sign-off

This report has been prepared based on monitoring data recorded during the period 01-Apr-2024 to 30-Jun-2024 and reflects, to the best of our knowledge, an accurate and complete summary of the environmental compliance status of FortTrace Dahej Integrated Manufacturing Complex for the reporting quarter.

Prepared By	Reviewed By	Approved By
A. Deshmukh	S. Kulkarni	R. Nair
Sr. Executive - EHS	Head - Environment, Health & Safety	Plant Manager
FortTrace Dahej Complex	FortTrace Dahej Complex	FortTrace Dahej Complex
Date: 08-Jul-2024	Date: 08-Jul-2024	Date: 08-Jul-2024

This document is maintained as part of the FortTrace Plant Operations EHS compliance record and is retained for a minimum period of five (5) years in accordance with the facility's Consent to Operate conditions and applicable statutory record-keeping requirements. For queries regarding this report, contact the EHS & Sustainability Department, FortTrace Dahej Integrated Manufacturing Complex.